

Why don't we have these AI data centers run on nuclear power?

Posted by u/Super-414 • 0 points • 19 comments

It seems that people are concerned about the energy costs, but wouldn't this eliminate that? It would also be much more compact than solar or wind.

Comments

u/FearlessFrank99 • 1 points • 2025-10-11 20:01 UTC

A lot of them would love to, but building new nuclear power plants is crazy expensive, takes years to do, and has a lot of opposition to them as people are afraid of them.

Small Nuclear Reactors (SMRs) are in development and if viable would be a good option, but they are still likely years away.

u/jayron32 • 1 points • 2025-10-10 17:08 UTC

They don't have dedicated power plants to run them. If your electric grid has a nuclear power plant hooked up to it, then data centers also on your grid are, in part, running on nuclear power.

u/5PQR • 1 points • 2025-10-10 16:41 UTC

[BBC: Google turns to nuclear to power AI data centres]
(<https://www.bbc.co.uk/news/articles/c748gn94k95o>)

u/notextinctyet • 1 points • 2025-10-10 16:30 UTC

That has nothing to do with AI. It's just asking, why don't we have more nuclear power? The main reason is that people were afraid of it and used their political power to stop construction of new nuclear plants for decades. And nowadays it's not cost competitive with renewables.

u/DiogenesKuon • 2 points • 2025-10-10 15:54 UTC

The power in the grid is mostly considered fungible, and AI data centers only take up a small amount of the total power we use. So the real question is should we increase our power capacity and should nuclear be one of the options we pursue. The answer to both of those is probable yes, but building new nuclear power in the US is politically difficult.

u/Teekno • 2 points • 2025-10-10 15:47 UTC

Some do, some don't, it depends on what power is available to them.

But with the advent of small modular reactors, I won't be surprised if we start to see new data center

construction with its own nuclear power source. SMRs have the potential to revolutionize how we do nuclear power.

u/Super-414 • 1 points • 2025-10-10 15:51 UTC

Without googling — what are SMRs?

u/Teekno • 2 points • 2025-10-10 15:54 UTC

Small modular (nuclear) reactors. One of the huge costs for nuclear plants is the fact that every single one of them is custom built, and any repairs or replacements also have to be custom built. A modular reactor is what it sounds like -- parts are standardized and can be mass produced, cutting costs. There's still a significant uphill regulatory battle to get any new reactors approved, though.

u/Super-414 • 1 points • 2025-10-10 16:01 UTC

Regulatory — so politics slowing us down again

u/Astramancer_ • 7 points • 2025-10-10 15:47 UTC

Honestly, if "We need more power to run our AI datacenters" is what gets the government off it's ass and start building new nuclear power plants, then so be it.

u/Concise_Pirate • 3 points • 2025-10-10 15:43 UTC

Many do. But nuclear is the most costly. Solar and wind are the cheapest.

u/DrHugh • 8 points • 2025-10-10 15:36 UTC

Nuclear power isn't free, you know.

Now, one could say that AI data centers must provide their own power, and can't be connected to the normal power grid. But I don't think the purveyors of AI would like that.

u/OZ-00MS_Goose • 2 points • 2025-10-10 15:35 UTC

I think eventually we will decide to just have data centers in the middle of nowhere with dedicated nuclear plants to serve them. Like the American Southwest has no natural disasters and tons of empty land

u/SLOBeachBoi • 1 points • 2025-10-10 16:36 UTC

Problem is you can't build a nuclear reactor away from a source of water and water sources tend to be where people are

u/56BPM • 1 points • 2025-10-10 15:59 UTC

Clear power is extraordinarily expensive to set up. And extraordinarily regulated. Geothermal seems to have better roi

u/rhomboidus • 2 points • 2025-10-10 15:52 UTC

> data centers in the middle of nowhere

One word on why that won't work: Latency.

u/Super-414 • 1 points • 2025-10-10 15:50 UTC

My thoughts exactly. I think they also need access to water for cooling, though. But exactly — if they become essential, make it the most efficient as possible.

u/Cliffy73 • 9 points • 2025-10-10 15:31 UTC

We do, some of 'em. They just run on whatever power is available where they are Nuclearization of the grid would be a generally good idea, but there are problems with it including waste storage and the fact that people freak out about it (not without reason), which means we are not utilizing nuclear power as much as we could.

u/ForScale • 4 points • 2025-10-10 15:30 UTC

Im gonna guess the costs involved would be greater.